

Asymmetric Capital Buffer Control

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1 Introduction

The question is how an asymmetric penalty on a capital shortfall shapes a bank's recapitalisation policy and the volatility of its capital buffer. The model treats recapitalisation as impulse control under a fixed issuance cost and a regulatory cap on risky exposure.

Table 1: Headlines.

ID	Headline	Model rows	Scope note
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2 Model

Table 2: Model results.

ID	Claim	Statement	Lean	Proof
M1	The deficit-to-surplus variance ratio of the discrete-time recursion is λ_V^4 , for every saturation scale.	$V_0 > 0, \theta > 0, \lambda_V > 0,$ $V(z) = V_0 \exp(-2 \log \lambda_V \tanh(z/\theta)) \Rightarrow$ $V(-z)/V(z) \rightarrow \lambda_V^4$ as $z \rightarrow \infty$.	<code>RateBased.Vresp_ratio_</code> <code>eq, RateBased.tendsto_</code> <code>tanh_atTop, RateBased.</code> <code>saturated_ratio_</code> <code>tendsto, RateBased.</code> <code>saturated_ratio_</code> <code>lambda4</code>	p. 2
M2	The solvency branch of the cap binds exactly up to the crossing point $\bar{x}$, and there the z -volatility is the constant κ_1^2 .	$0 < a_1 < a_3, a_2 < 1 \Rightarrow$ $\bar{x} - 1 = a_1(1 - a_2)/(a_3 - a_1) > 0,$ $u(x) = (x - 1)/a_1$ for $x \leq \bar{x}, u(x) = (x - a_2)/a_3$ for $x \geq \bar{x}$. If also $q > 0, 1 < x \leq \bar{x}$ and $\pi(x)x = u(x)$, then $\zeta(x) = \kappa_1^2$.	<code>CapGeometry.</code> <code>hasDerivAt_zmap,</code> <code>CapGeometry.zvol2_eq,</code> <code>CapGeometry.xbar_sub_</code> <code>one, CapGeometry.one_</code> <code>lt_xbar, CapGeometry.</code> <code>solvency_binds,</code> <code>CapGeometry.liquidity_</code> <code>binds, CapGeometry.</code> <code>zvol2_solvency</code>	p. 3
M3	For any exposure policy that binds the cap near an end of the state space, the z -volatility tends to κ_1^2 at the distress boundary and to κ_3^2 deep in surplus.	$q > 0, 0 < a_1 < a_3, a_2 < 1$. If $\pi(x)x = u(x)$ for all x near 1^+ then $\zeta(x) \rightarrow \kappa_1^2$ as $x \rightarrow 1^+$. If $\pi(x)x = u(x)$ for all large x then $\zeta(x) \rightarrow \kappa_3^2$ as $x \rightarrow \infty$. Both fail without $a_1 < a_3$ or without binding.	<code>CapGeometry.deficit_</code> <code>limit, CapGeometry.</code> <code>surplus_limit,</code> <code>CapGeometry.variance_</code> <code>ratio_is_cap_geometry,</code> <code>CapGeometry.control_</code> <code>surplus_limit_needs_</code> <code>a1_lt_a3, CapGeometry.</code> <code>control_constancy_</code> <code>needs_binding</code>	p. 3

3 Literature

Table 3: Papers read.

ID	Paper	What we rely on	Quote	Page	Rows
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Bayraktar, Chevalier, Ly Vath and Wang (arXiv:2603.14557), Barberis, Huang and Santos (2001) and Engle and Siriwardane (2018) are named in Table 5 but have not been read, and every claim made about them is unverified.

4 Conclusions

Table 4 holds the conclusions that rest on model rows and on papers read in full. Table 5 holds every claim that a result is new and every other claim still waiting, and each of its rows names what it waits on.

Table 4: Conclusions.

ID	Statement	Rows
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Table 5: Pending conclusions.

ID	Statement	Rows
C1	Wherever the optimal policy binds the regulatory cap near both ends of the state space, the saturation limits of the z -volatility are κ_1^2 and κ_3^2 . Their ratio involves $(\sigma, \sigma_L, a_1, a_3, c)$ only, so read as the variance ratio λ_V^4 it carries no information about λ_S . <i>Pending</i> : the novelty reading of Bayraktar, Chevalier, Ly Vath and Wang (arXiv:2603.14557), Barberis, Huang and Santos (2001) and Engle and Siriwardane (2018); and the binding hypothesis, which rests on numerical evidence only (Appendix B).	M1, M2, M3

A Proofs

Notation. In discrete time $V(z)$ is the conditional variance of the shock at log buffer ratio z , with induced variance asymmetry $\lambda_V > 0$, saturation scale $\theta > 0$ and level $V_0 > 0$. In continuous time $x \in (1, \infty)$ is the ratio of assets to liabilities, $q > 0$ the ratio requirement, and $z(x) = \log((x-1)/q)$. The risky exposure is $\pi(x)$, and the regulatory cap gives the capped exposure $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with crossing point $\bar{x} = (a_3 - a_1 a_2)/(a_3 - a_1)$. The diffusion of x is $\sigma^2(x, \pi) = \pi^2 \sigma^2 x^2 + 2\pi c \sigma \sigma_L x(1-x) + \sigma_L^2 (1-x)^2$, with c the correlation between the risky return and the liability shock. The z -volatility is $\zeta(x) = z'(x)^2 \sigma^2(x, \pi(x))$, the exposure ratio is $\omega(x) = \pi(x)x/(x-1)$, and $\kappa_i^2 = \sigma^2/a_i^2 - 2c\sigma\sigma_L/a_i + \sigma_L^2$ for $i \in \{1, 3\}$.

Proof of M1

Lean: RateBased.Vresp_ratio_eq, RateBased.tendsto_tanh_atTop, RateBased.saturated_ratio_tendsto, RateBased.saturated_ratio_lambda4.

For every z ,

$$\begin{aligned} \frac{V(-z)}{V(z)} &= \exp(4 \log \lambda_V \tanh(z/\theta)). \\ \tanh y &= 1 - \frac{2}{e^{2y} + 1} \rightarrow 1 \quad (y \rightarrow \infty), \quad z/\theta \rightarrow \infty \quad (z \rightarrow \infty, \theta > 0). \\ \exp(4 \log \lambda_V \tanh(z/\theta)) &\rightarrow \exp(4 \log \lambda_V) = \lambda_V^4. \end{aligned}$$

θ enters only through $z/\theta \rightarrow \infty$, so the limit is free of θ .

□

Proof of M2

Lean: CapGeometry.hasDerivAt_zmap, CapGeometry.zvol2_eq, CapGeometry.xbar_sub_one, CapGeometry.one_lt_xbar, CapGeometry.solvency_binds, CapGeometry.liquidity_binds, CapGeometry.zvol2_solvency.

$$\frac{x-1}{a_1} \leq \frac{x-a_2}{a_3} \iff (a_3-a_1)x \leq a_3-a_1a_2 \iff x \leq \bar{x}, \quad \bar{x}-1 = \frac{a_1(1-a_2)}{a_3-a_1} > 0.$$

$$z'(x) = \frac{1}{x-1}, \quad \zeta(x) = \omega(x)^2 \sigma^2 - 2\omega(x)c\sigma\sigma_L + \sigma_L^2.$$

$$1 < x \leq \bar{x}, \pi(x)x = u(x) \Rightarrow \omega(x) = \frac{1}{a_1} \Rightarrow \zeta(x) = \kappa_1^2.$$

□

Proof of M3

Lean: CapGeometry.deficit_limit, CapGeometry.surplus_limit, CapGeometry.variance_ratio_is_cap_geometry, CapGeometry.control_surplus_limit_needs_a1_lt_a3, CapGeometry.control_constancy_needs_binding.

Near 1^+ , $\bar{x} > 1$ by M2, so $1 < x < \bar{x}$ eventually and $\zeta(x) = \kappa_1^2$ there. For $x \geq \bar{x}$,

$$\omega(x) = \frac{x-a_2}{a_3(x-1)} = \frac{1}{a_3} + \frac{1-a_2}{a_3}(x-1)^{-1} \rightarrow \frac{1}{a_3},$$

$$\zeta(x) = \omega(x)^2 \sigma^2 - 2\omega(x)c\sigma\sigma_L + \sigma_L^2 \rightarrow \kappa_3^2.$$

Controls, at $\sigma = 1$, $\sigma_L = c = 0$, $q = 1$.

$$a_1 = \frac{1}{2}, a_2 = 0, a_3 = \frac{1}{4} \Rightarrow u(x) = 2(x-1), \zeta \equiv 4 \neq 16 = \kappa_3^2.$$

$$\pi \equiv 0 \Rightarrow \zeta(2) = 0 \neq 4 = \kappa_1^2.$$

□

B Evidence and verification

Toolchain. Lean v4.32.0-rc1 with Mathlib at tag v4.32.0-rc1, commit 360da6fa66c1, pinned in lean/mathlib/lake-manifest.json. Every file under lean/mathlib/ is a build root. A theorem passes the axiom audit when it depends on no axiom outside propext, Classical.choice and Quot.sound.

Checks. verify.sh runs checks/verify_manuscript.py, checks/verify_mathlib.py and checks/verify_docs.py, and writes their output to VERIFICATION.md. lit/verify_lit.sh runs the suite of every literature record.

What Lean does not carry. Two inputs enter the theorems as hypotheses or definitions. The z-volatility ζ is defined as $z'(x)^2 \sigma^2(x, \pi(x))$, and that this is the diffusion coefficient of $z(X_t)$ is Itô's formula, which is not proved (M2, M3). That the optimal policy binds the cap near both ends of the state space is a hypothesis of M3. PROOFS_v2.tex supports it only by the numerical solve, which is evidence and not proof (C1).

Illustrations. No row rests on SymPy or on numerical sampling.

C Terminology

Term	Symbol	Meaning
Induced variance asymmetry	λ_V	The asymmetry read off the variance ratio, distinct from λ_S .
Variance ratio	λ_V^4	The ratio of conditional variances deep in deficit to deep in surplus (M1).
Saturation scale	θ	How sharply the conditional variance moves between its deficit and surplus levels.
Asymmetry parameter	λ_S	The weight on shortfall below the reference in the objective.
Regulatory cap	$u(x)$	The capped exposure $\min\{(x-1)/a_1, (x-a_2)/a_3\}$.
Cap geometry	a_1, a_2, a_3	The parameters of the regulatory cap.
Solvency and liquidity regime	$x \leq \bar{x}, x \geq \bar{x}$	The ranges where the first and the second branch of u bind (M2).
z -volatility	$\zeta(x)$	The squared diffusion coefficient of z (Appendix B).
Saturation limits	κ_1^2, κ_3^2	The limits of ζ at the distress boundary and deep in surplus (M3).

D Primitives and scope

The regulatory cap. The capped exposure is $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with $0 < a_1 < a_3$ and $a_2 < 1$, taken from PROOFS_v2.tex. Under these restrictions the solvency branch binds near the distress boundary and the liquidity branch deep in surplus (M2). With $a_1 \geq a_3$ the surplus limit of M3 fails, by its control. Rows M2, M3 and C1 depend on it.

The variance ratio in continuous time. PROOFS_v2.tex reads the discrete-time λ_V^4 of M1 in the continuous-time model as κ_1^2/κ_3^2 , identifying the conditional variance $V(z)$ deep in deficit and deep in surplus with the two saturation limits of ζ . This is a definition, not a result, and no theorem connects the discrete-time recursion to the continuous-time diffusion. C1 depends on it.

E Refuted conjectures